

CS4268

Quantum Computing

Midterm Solutions

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Question 1.

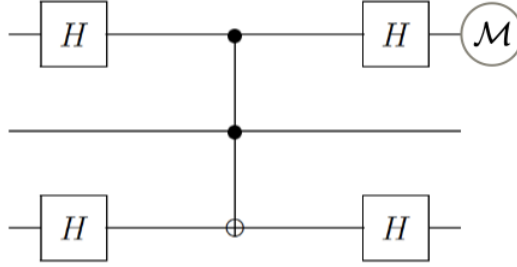


Figure 1

In analyzing the circuit in Figure 1, for brevity we shall represent the action of various gates on the $|0\rangle, |1\rangle$ states in the following manner:

$$\begin{aligned}
 |x\rangle &\xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + (-1)^x|1\rangle) \\
 \frac{1}{\sqrt{2}}(|0\rangle + (-1)^x|1\rangle) &\xrightarrow{H} |x\rangle \\
 |x\rangle|y\rangle|z\rangle &\xrightarrow{\text{Toff}} |x\rangle|y\rangle|z \oplus (x \wedge y)\rangle
 \end{aligned}$$

when $x, y, z \in \{0, 1\}$. It can be verified by substitution that these are the correct actions.

For clarity, we shall number the qubits 1 to 3 from top to bottom in Figure 1 and we shall denote operations on specific qubits with the corresponding numbers in the suffix. Now we determine the action of the circuit as:

$$\begin{aligned}
 \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle) &\xrightarrow{H_1 \otimes H_3} \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) + \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)|1\rangle \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \right) \\
 &= \frac{1}{2\sqrt{2}} (|000\rangle + |001\rangle + |010\rangle - |011\rangle + |100\rangle + |101\rangle - |110\rangle + |111\rangle) \\
 &\xrightarrow{\text{Toff}_{123}} = \frac{1}{2\sqrt{2}} (|000\rangle + |001\rangle + |010\rangle - |011\rangle + |100\rangle + |101\rangle + |110\rangle - |111\rangle) \\
 &\xrightarrow{H_1 \otimes 1} = \frac{1}{2} (|000\rangle + |001\rangle + |010\rangle - |011\rangle)
 \end{aligned}$$

$$\xrightarrow{\mathbb{1} \otimes H_3} \frac{1}{\sqrt{2}} (|000\rangle + |011\rangle).$$

Hence, the circuit in Figure 1 after acting on $\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ gives 0 with probability 1.

Question 2.

The circuit in Figure 2 performs the transformation $x \in \{0, 1\}^n, y \in \{0, 1\}^n, b \in \{0, 1\}$:

$$|x\rangle|y\rangle|b\rangle \rightarrow |x\rangle|y\rangle|b \oplus z\rangle$$

where

$$z = \left(\sum_{i=1}^n x_i y_i \right) \mod 2.$$

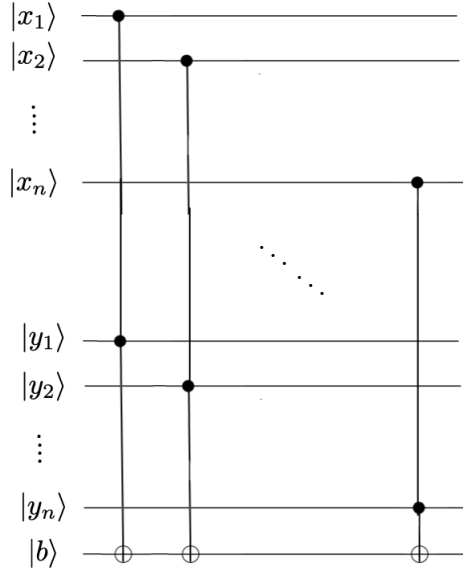


Figure 2

In analyzing the action of the circuit, we shall use subscripts for the gates to denote which qubit they are acting on. For example, $\text{Toff}_{1,2,3}$ denotes the Toffoli gate acting on the first three qubits from the top, with the convention being that the first, second number in the subscript denotes the control qubit. We shall also omit tensor factors of identity in the gates. We can see the action of the circuit on the initial state as follows

$$\begin{aligned} |x\rangle|y\rangle|b\rangle &\xrightarrow{\text{Toff}_{1,n+1,2n+1}} |x\rangle|b \oplus (x_1 y_1)\rangle \\ &\xrightarrow{\text{Toff}_{2,n+2,2n+1}} |x\rangle|y\rangle|b \oplus (x_1 y_1) \oplus (x_2 y_2)\rangle \\ &\vdots \end{aligned}$$

$$\xrightarrow{\text{Toff}_{n,n+n,2n+1}} |x\rangle|y\rangle|b \oplus ((x_1y_1) \oplus (x_2y_2) \oplus \dots \oplus (x_ny_n))\rangle$$

$$= |x\rangle|y\rangle|b \oplus z\rangle$$

where

$$z = \left(\sum_{i=1}^n x_i y_i \right) \mod 2.$$

Question 3.

The circuit in Figure 3 lets us determine the unitary U by one single use of U as black box.

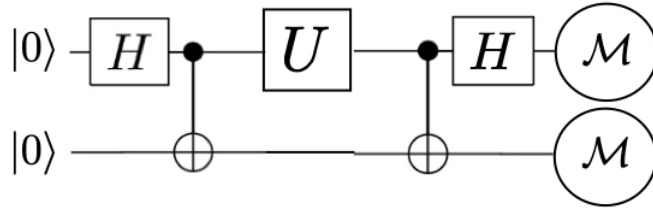


Figure 3

Suppose $U = \sigma_x$, the outcome of the circuit will be

$$\begin{aligned} |0\rangle|0\rangle &\xrightarrow{H \otimes 1} \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) |0\rangle \\ &\xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \\ &\xrightarrow{\sigma_x \otimes 1} \frac{1}{\sqrt{2}} (|10\rangle + |01\rangle) \\ &\xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}} (|11\rangle + |01\rangle) \\ &\xrightarrow{H \otimes 1} |0\rangle|1\rangle. \end{aligned}$$

If $U = \sigma_y$, the outcome of the circuit will be

$$\begin{aligned} |0\rangle|0\rangle &\xrightarrow{H \otimes 1} \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) |0\rangle \\ &\xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \\ &\xrightarrow{\sigma_y \otimes 1} \frac{i}{\sqrt{2}} (|10\rangle - |01\rangle) \\ &\xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}} (|11\rangle - |01\rangle) \\ &\xrightarrow{H \otimes 1} |1\rangle|1\rangle. \end{aligned}$$

Finally if $U = \sigma_z$, the outcome of the circuit will be

$$\begin{aligned}
 |0\rangle|0\rangle &\xrightarrow{H\otimes\mathbb{1}} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle \\
 &\xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \\
 &\xrightarrow{\sigma_z\otimes\mathbb{1}} \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \\
 &\xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle) \\
 &\xrightarrow{H\otimes\mathbb{1}} |1\rangle|0\rangle.
 \end{aligned}$$

Question 4.

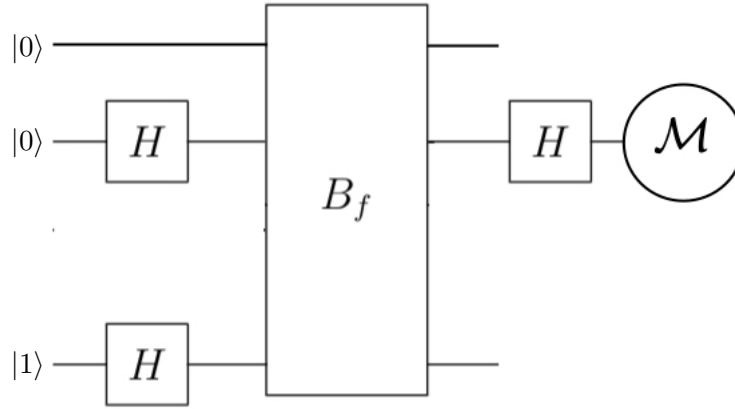


Figure 4

(a) The action of the circuit 4 is given by

$$\begin{aligned}
 |00\rangle|1\rangle &\xrightarrow{\mathbb{1}\otimes H\otimes H} \frac{1}{2}((|00\rangle + |01\rangle)(|0\rangle - |1\rangle)) \\
 &\xrightarrow{B_f} \frac{1}{2}((-1)^{f(00)}|00\rangle + (-1)^{f(01)}|01\rangle)(|0\rangle - |1\rangle) \\
 &\xrightarrow{\mathbb{1}\otimes H\otimes\mathbb{1}} \frac{(-1)^{f(00)}}{\sqrt{2}}(|0\rangle|f(00) \oplus f(01)\rangle)(|0\rangle - |1\rangle).
 \end{aligned}$$

Thus the measurement will output $(f(00) \oplus f(01))$.

(b) If $(f(00) \oplus f(01)) = 0$, then query and output $f(00)$; else query and output $f(10)$. This works because if $(f(00) \oplus f(01)) = 0$ then $f(00) = f(01)$ and hence these bits are the majority; and if $(f(00) \oplus f(01)) = 1$ then $f(00) \neq f(01)$ and hence $f(10)$ determines the majority. Thus exactly 2 queries suffice to compute majority of $\{f(00), f(01), f(10)\}$.